Bivariate Normal : Practice.

Problem. Midterm and Final.

Midterm and final scores in a large class have an (approximately) bivariate normal dist'n w/ parameters:

	mean	sd
midterm	65	18
final	60	20

correlation : 0.76

Q: What is the "estimated" mean final score of the students who were above the mean on the midterm?

→: Let U be the midterm score, and let V be the final score.

Our task is to find:

$$\boxed{\mathbb{E}[V \mid U > \mu_U] = ?}$$

Let X and Z be U and V in std units, resp.

Our ancillary task is to find:

$$\mathbb{E}[Z \mid X > 0] = ?$$

$$\mathbb{E}[Z \mid X>0] = \frac{\mathbb{E}[Z \cdot \mathbb{I}_{[X>0]}]}{\mathbb{P}[X>0]} = \cdots$$

by def'n

$$\boxed{\mathbb{E}[Z \mid X>0]} = \int_{\cancel{-\infty}\ 0}^{\infty} \boxed{\mathbb{E}[Z \mid X=x]} \cdot \boxed{f_X(x \mid X>0)}\, dx$$

Law of Total Probability

ρx

$\boxed{Z \mid X=x \sim \text{Normal}(\rho x, \text{var} = 1-\rho^2)}$

for $x>0$: $f_X(x \mid X>0)\,dx = \mathbb{P}[X \in (x, x+dx) \mid X>0]$

$$= \frac{\mathbb{P}[X \in (x, x+dx),\ X>0]}{\mathbb{P}[X>0]}$$

$$= \frac{\mathbb{P}[X \in (x, x+dx)]}{\frac{1}{2}} = 2 \cdot f_X(x)$$

$$= 2 \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

$$\mathbb{E}[Z \mid X>0] = \int_0^\infty \rho x \cdot 2 \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$$

$$= \frac{2\rho}{\sqrt{2\pi}} \int_0^\infty x e^{-\frac{x^2}{2}} dx = \left[u = \frac{x^2}{2} \Rightarrow du = \frac{1}{\cancel{2}} \cancel{2} x\, dx = x\, dx \right]$$

$$= \frac{2\rho}{\sqrt{2\pi}} \underbrace{\int_0^\infty e^{-u} du}_{=1} = \frac{2\rho}{\sqrt{2\pi}}$$

<u>In this problem:</u>

$$\frac{1.52}{\sqrt{2\pi}} \approx 0.6063923$$

$\Rightarrow$ <u>Our final answer:</u> $60 + 20 \cdot \frac{1.52}{\sqrt{2\pi}} \approx \underline{72.12785}$

$\square$

The University of Texas at Austin
In-Class Work 3

M339G Predictive Analytics

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Bivariate Normal Distribution.

Problem 3.1. (15 points) The extremely popular *Fit Fathers* exercise club team enrolled in a fitness regimen. Here is a summary of their pre- and post-regimen weights (in lbs).

	mean	standard deviation
PRE	220	20
POST	185	10

Assume that the data are modeled as bivariate normal with the correlation coefficient equal to 0.8. Of the fathers who weighed ~~X~~ exactly 260 pounds before the fitness regimen, about what percentage weighed above average after the regimen?

→: $\boxed{U \ldots \text{ pre-regimen weight}}$ $\longleftrightarrow$ X in std units

V... post-regimen weight $\longleftrightarrow$ Y in std units

Conditioning on $U = 260$

$\Longleftrightarrow$

Conditioning on $X = \dfrac{260-220}{20} = 2$

Looking for

$$\mathbb{P}[V > 185 \mid U = 260] = \mathbb{P}\left[\frac{V-185}{10} > \frac{185-185}{10} \,\middle|\, X = 2\right]$$

$$= \mathbb{P}[Y > 0 \mid X = 2]$$

$Y \sim \text{Normal}(\rho x, \text{var} = 1-\rho^2)$

$$\mathbb{P}\left[\frac{Y-\rho x}{\sqrt{1-\rho^2}} > \frac{0-\rho x}{\sqrt{1-\rho^2}}\right] = \mathbb{P}\left[Z > \frac{-\rho x}{\sqrt{1-\rho^2}}\right] =$$

$\sim N(0,1)$

$$= \Phi\left(\frac{\rho x}{\sqrt{1-\rho^2}}\right) = \Phi\left(\frac{0.8 \cdot 2}{\sqrt{1-0.64}}\right) = \Phi\left(\frac{1.6}{0.6}\right) = \Phi\left(\frac{8}{3}\right)$$